

Supporting Information for

Molecular Alloys, Linking Organometallics with Intermetallic Hume-Rothery Phases: The Highly Coordinated Transition Metal Compounds
 $[M(ZnR)_n]$ ($n \geq 8$) Containing Organo-Zinc Ligands.

by

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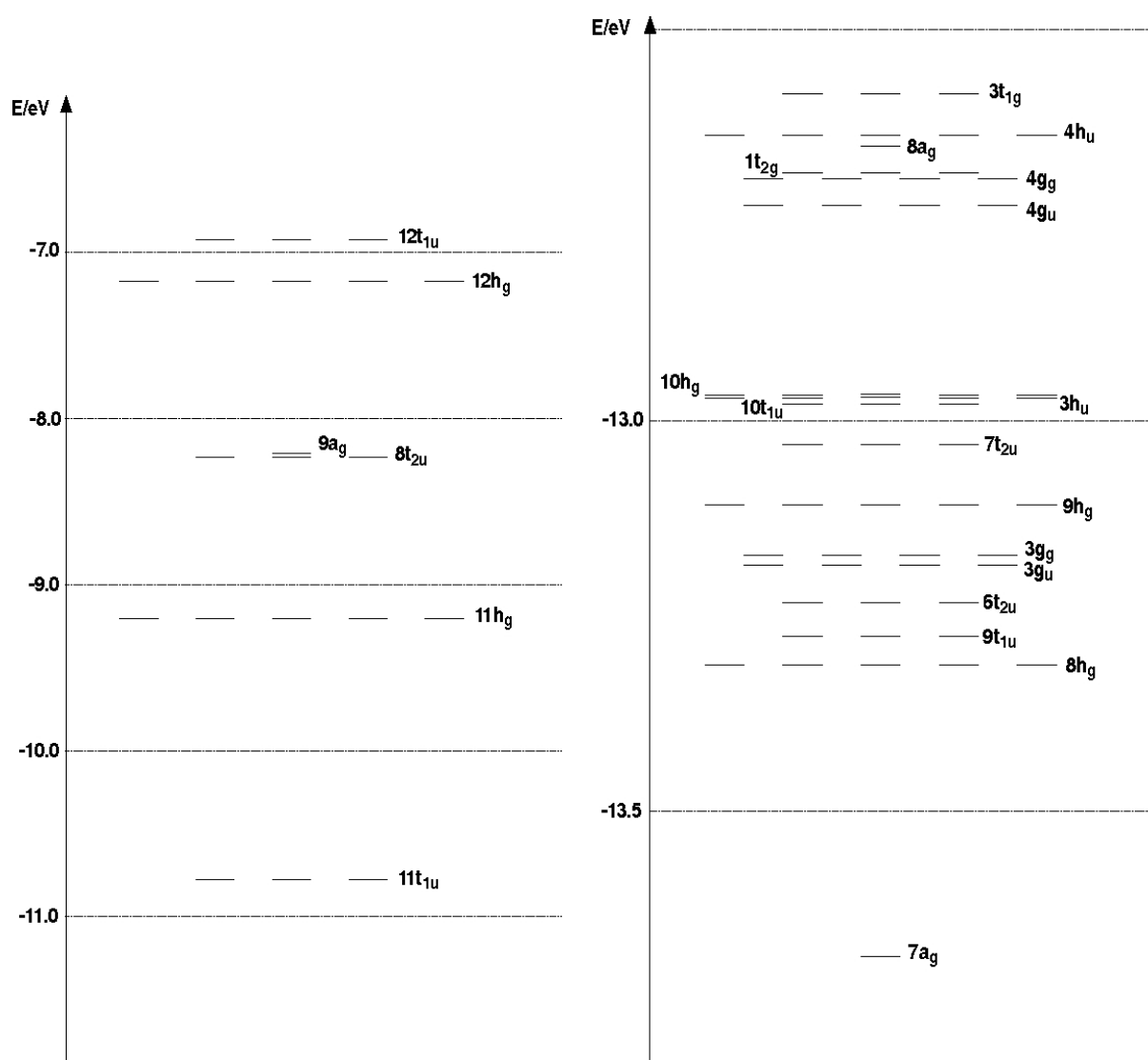


Figure S1. Energy Levels of the complete set of occupied valence orbitals of **1H**.

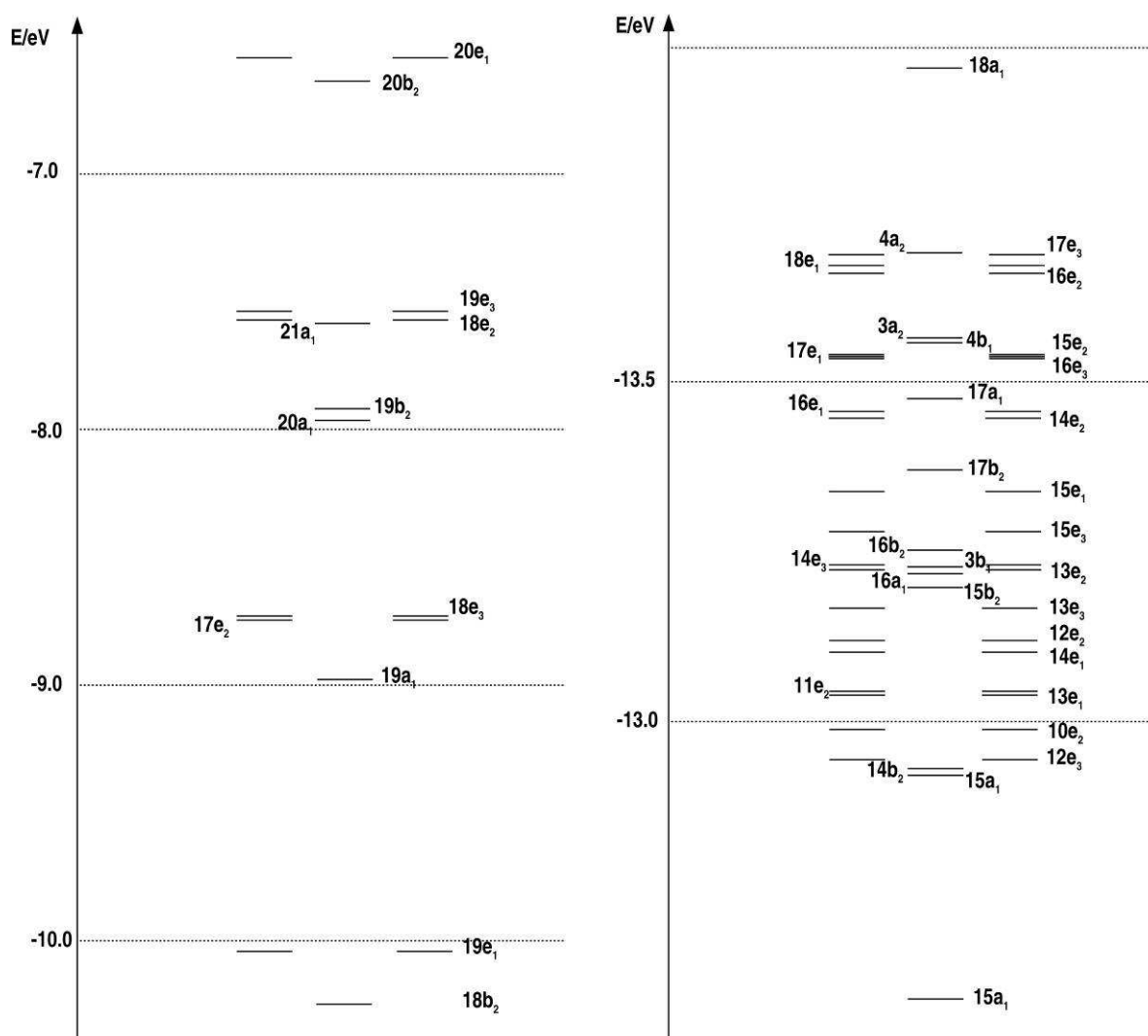


Figure S2. Energy Levels of the complete set of occupied valence orbitals of **5H**.

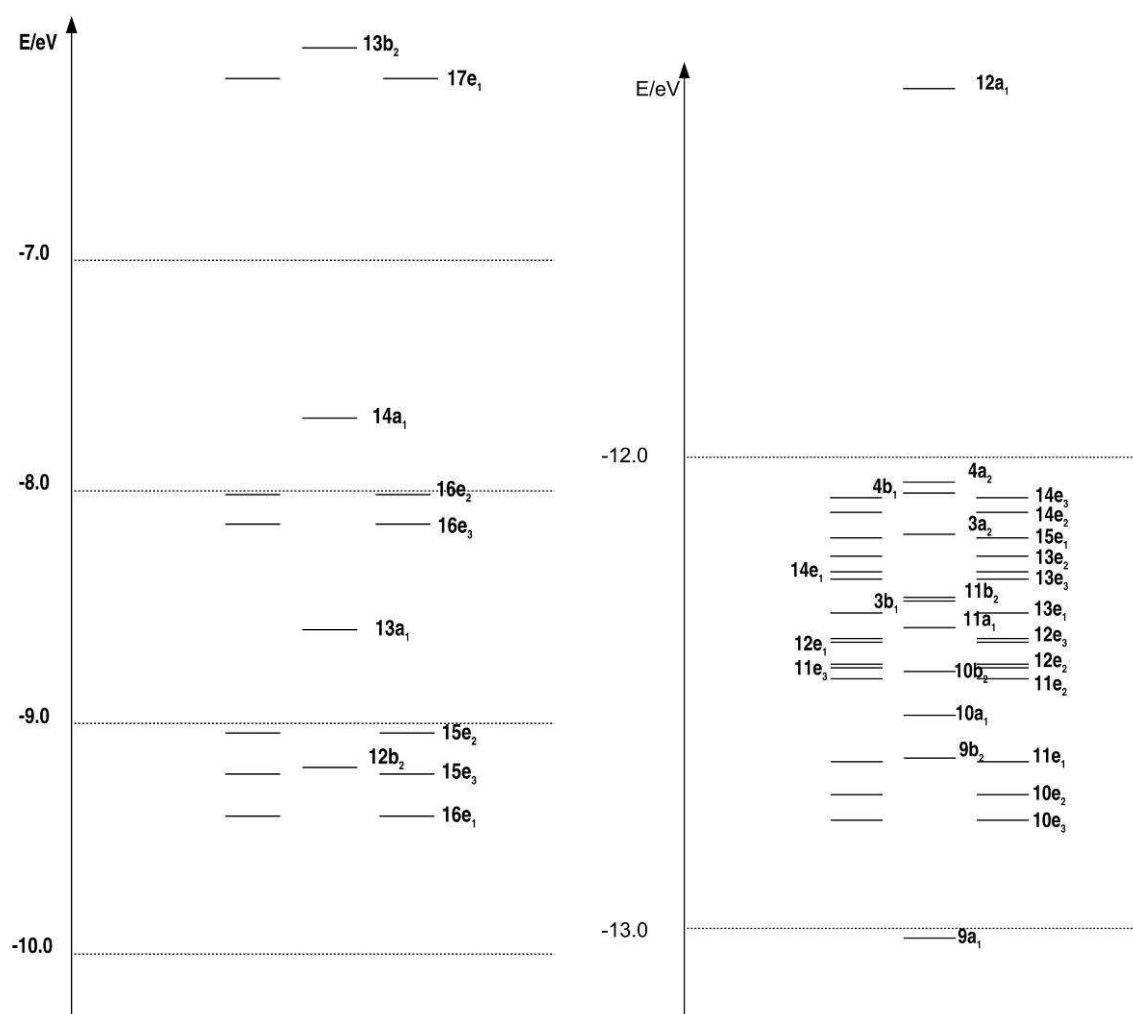


Figure S3. Energy Levels of the complete set of occupied valence orbitals of **10H**.

Bonding Situation in [Rh(ZnH)₉] (**7H**) in comparison with [Mo(ZnH)₁₂] (**1H**)

The 29e'(HOMO) of [Rh(ZnH)₉] (**7H**) has large coefficients from the p_x(Rh) and p_y(Rh) AOs with small contributions from d_{xy}(Rh) and d_{x²-y²}(Rh) orbitals. The 16a₂'(HOMO-1) has only contributions from the p_z(Rh) AO. The nearly degenerate orbitals are comparable to the 12t_{1u}(HOMO) of [Ru(ZnH)₁₀] (**5H**) (Figure 5). Below is the 22a₁'(HOMO-2) orbital which has significant contributions from s(Rh) with some mixing of d_{z²}(Rh) AOs (Figure S4). The next lower lying orbitals can be grouped in the set of five orbitals 28e'(HOMO-3), 21a'(HOMO-4) and 20e''(HOMO-5). They are easily identified as orbitals which have larger contributions from the valence d(Rh) AOs. The latter sets of orbitals are similar to the 12h_g(HOMO-1) and 9a_g(HOMO-2) of [Ru(ZnH)₁₀] (Figure 5) but with reversed ordering. Figure S4 shows that there are two more sets of orbitals in **7H** which consists of five MOs that have significant coefficients at the d(Rh) AOs (20a₁', 27e', 19e'' and 15a₁', 18e', 12e''). Furthermore there are two lower lying sets of nearly triply degenerate orbitals (16a₂'', 27e' and 11a₂'', 19e') which have small coefficients from the p(Rh) AOs. Finally there are two more MOs which have large contributions from the valence s(Rh) AO (19a₁' and 14a₁'). The EDA results for **7H** are given in Table S4 (below). The calculations were carried out using the closed-shell charged fragments Rh⁻ (s⁰d¹⁰) and (ZnH)₉⁺ in the singlet states and the open-shell neutral fragments Rh (s⁰d⁹) and (ZnH)₉ in the doublet states. Both EDA calculations suggest that the electrostatic bonding is stronger than the orbital (covalent) attraction. The absolute value for the orbital term ΔE_{Orb} of the former calculation (-172.8 kcal/mol) is clearly smaller than in the latter calculations (-241.0 kcal/mol) suggesting that the bonding in **7H** should be discussed in terms of Rh⁻-(ZnH)₉⁺ interactions. The EDA results show that the largest contributions to ΔE_{Orb} come from ΔE(e₁'') (d_{xz}, d_{yz}) while the ΔE(a₂'') (p_z) contribution is negligible. This means that the large ΔE(e₁') (p_x, p_y, d_{xy}, d_{x²-y²}) value comes also mainly from the d(Rh) AOs and that the large contribution of ΔE(a₁') (s, d_{z²}) comes also mainly from the d_{z²}(Rh) AO.

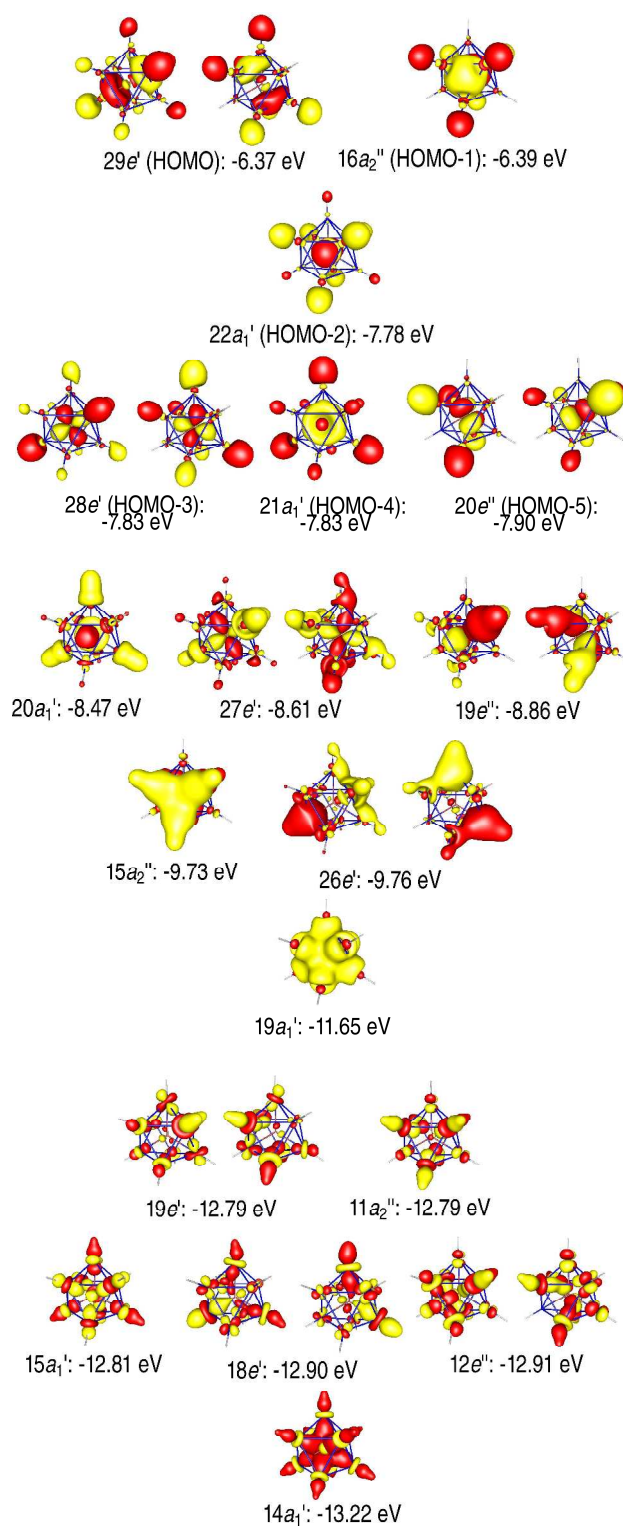


Figure S4. Most important occupied valence MOs of **7H** which have coefficients at Rh. The energy levels of the complete set of 63 occupied valence orbitals of **7H** are shown in Figure S5.

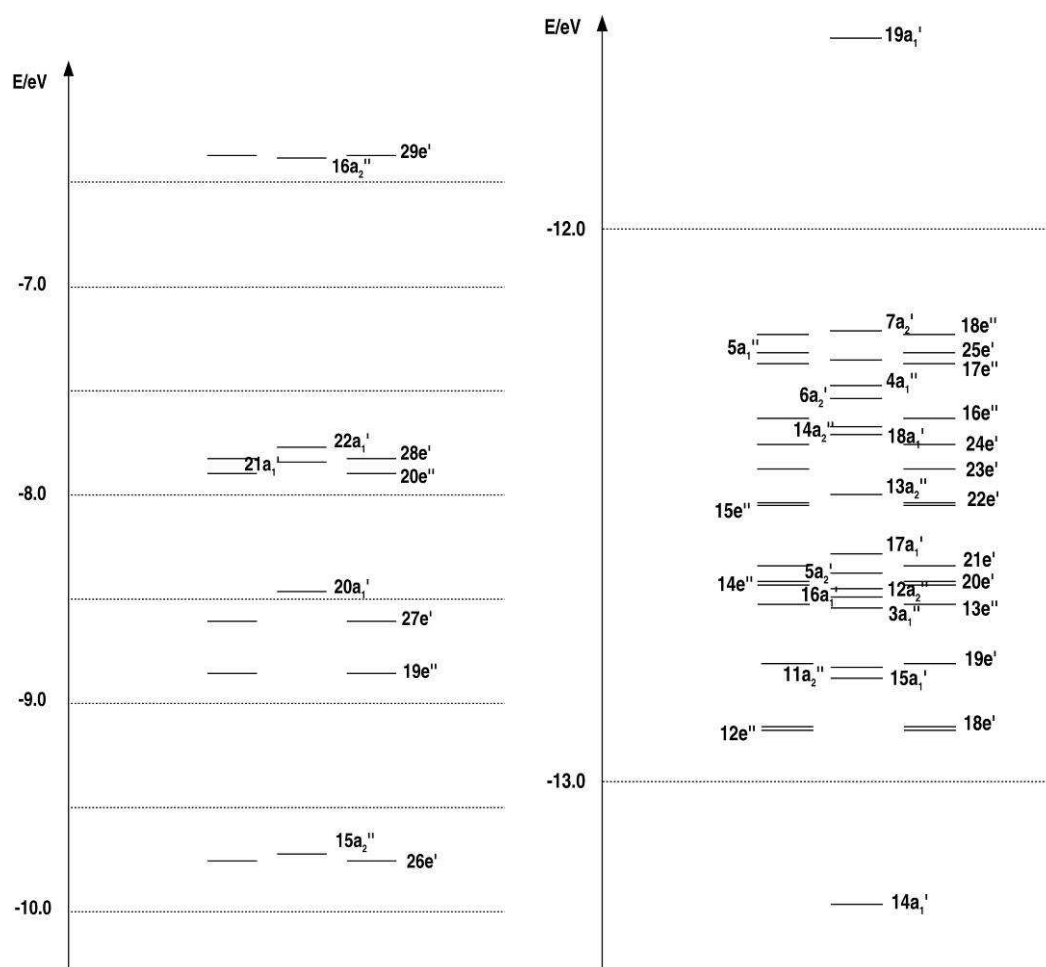


Figure S5. Energy Levels of the complete set of occupied valence orbitals of **7H**.

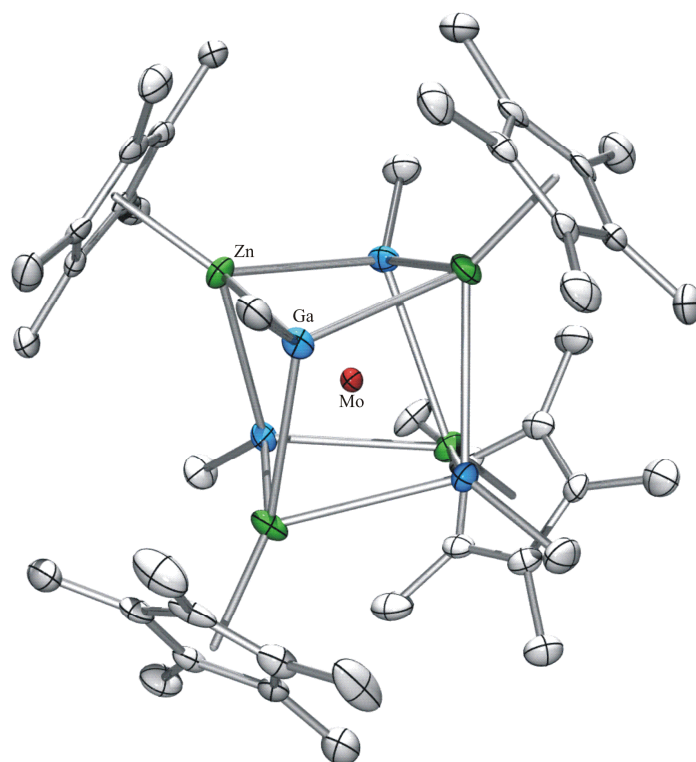


Figure S6. *Proposed* molecular structure of $[\text{MoGa}_4\text{Zn}_4\text{Me}_4\text{Cp}^*_4]$ (**3**) (thermal ellipsoids are shown at the 50% probability level, hydrogen atoms have been omitted for clarity). The distinction between the Zn and the Ga positions and the distribution of the Me and Cp* substituents over the Ga and Zn positions could not be unambiguously verified by the data refinement. However, model calculations on the DFT level of theory with permutation of the Zn and Ga positions suggest the particular structural assignment which was the basis of the X-ray data refinement. Thus, the coordination of the Cp* group is preferred at the Zn.

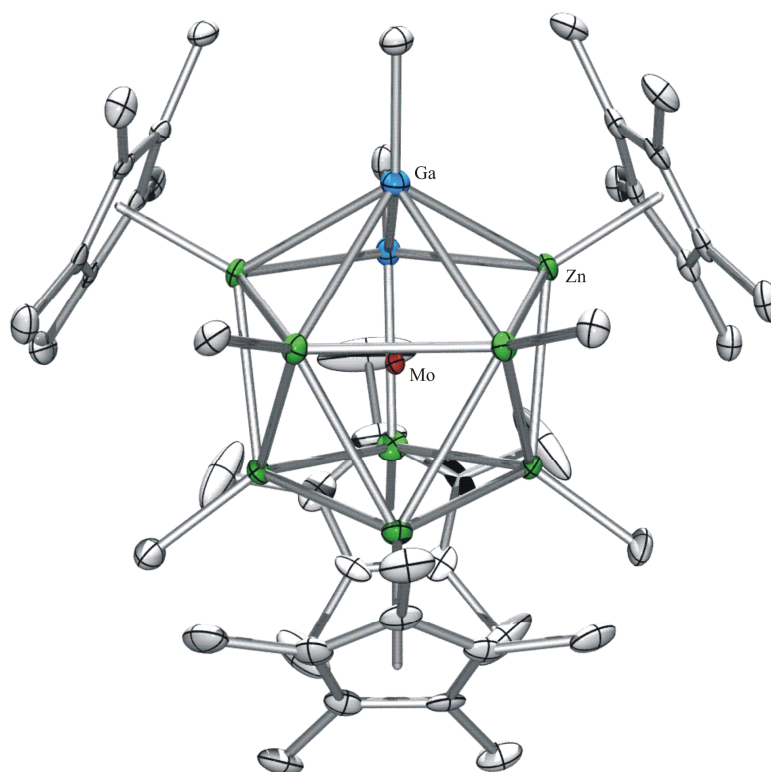


Figure S7. *Proposed* molecular structure of **4** (thermal ellipsoids are shown at the 30 % probability level, hydrogen atoms have been omitted for clarity). See the comment to Figure S6 and the general discussion of compounds **3** and **4** in the main text.

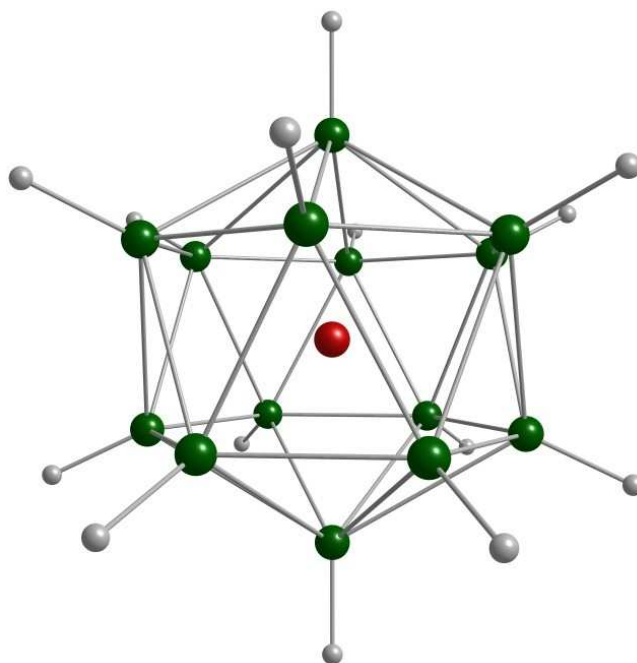


Figure S8. *Calculated* molecular structure of $[\text{Zr}(\text{ZnH})_{14}]$. The calculations show that $[\text{Zr}(\text{ZnH})_{14}]$ is a minimum on the potential energy surface which has D_{6d} symmetry. Thus, there is good reason to expect that a substituted homologue of $[\text{Zr}(\text{ZnH})_{14}]$ may become isolated as a stable species in a condensed phase, provided that suitable educts are found. Analysis of the bonding in $[\text{Zr}(\text{ZnH})_{14}]$ reveals an interesting difference with regard to $[\text{Mo}(\text{ZnH})_{12}]$. There are 14 bond paths for the Zr-Zn interactions but there are also Zn-Zn bond paths between the Zinc atoms of the planar Zn_6 moieties in $[\text{Zr}(\text{ZnH})_{14}]$ which means that there is stronger Zn-Zn bonding in the latter compound than in $[\text{Mo}(\text{ZnH})_{12}]$. This is reasonable because in the latter compound there are six electron pairs for Mo-Zn bonding and only three electron pairs for Z-Zn bonding while in $[\text{Zr}(\text{ZnH})_{14}]$ there are only four electron pairs for Zr-Zn bonding and five electron pairs for Zn-Zn bonding. A more detailed report about $[\text{Zr}(\text{ZnH})_{14}]$ and related compounds will be reported in a future work.

Table S1. Experimental details for the crystal structure refinements of compounds **1-12**.

Crystal parameters	1	2	3	4
Empirical formula	C ₃₉ H ₇₂ MoZn ₁₂	C ₄₀ H ₃₀ MoZn ₁₂	C ₄₄ H ₇₂ Ga ₄ MoZn ₄	C ₅₂ H ₈₄ Ga ₂ MoZn ₈
Molecular weight [g/mol]	1421.35	1391.02	1237.32	1467.53
Temperature [K]	113(2)	113(2)	113(2)	113(2)
Wavelength Mo-K α [Å]	0.71073	0.71073	0.71073	0.71073
Crystal size [mm]	0.2 × 0.1 × 0.1	0.35 × 0.25 × 0.1	0.25 × 0.2 × 0.1	0.35 × 0.25 × 0.15
Crystal system	Orthorhombic	Monoclinic	Monoclinic	Monoclinic
space group	Pnma	P2(1)/c	P2(1)/n	P2(1)/n
<i>a</i> (Å)	25.0480(7)	12.1242(8)	12.745(3)	17.3477(8)
<i>b</i> (Å)	14.7952(4)	21.3988(11)	23.559(5)	17.5178(9)
<i>c</i> (Å)	15.1100(4)	23.125(3)	18.921(5)	18.8957(9)
α (°)	90	90	90	90
β (°)	90	90	93.81(2)	94.774(4)
γ (°)	90	90	90	90
Cell volume [Å ³]	5599.6(3)	5999.7(9)	5693.9(13)	5722.3(5)
<i>Z</i>	4	4	4	4
Density ρ_{calc} [g cm ⁻³]	1.686	1.540	1.450	1.703
Absorption coefficient μ [mm ⁻¹]	5.273	4.920	3.767	4.455
<i>F</i> (000)	2832	2688	2488	2960
θ range for data collection [°]	2.79 - 27.57	2.69 - 27.63	3.05 -25.00	2.86 - 25.00
Index ranges	-29 ≤ <i>h</i> ≤ 32, -19 ≤ <i>k</i> ≤ 15, -19 ≤ <i>l</i> ≤ 19	-15 ≤ <i>h</i> ≤ 11, -27 ≤ <i>k</i> ≤ 27, -30 ≤ <i>l</i> ≤ 30	-15 ≤ <i>h</i> ≤ 15, -28 ≤ <i>k</i> ≤ 19, -22 ≤ <i>l</i> ≤ 22	-20 ≤ <i>h</i> ≤ 16, -20 ≤ <i>k</i> ≤ 20, -22 ≤ <i>l</i> ≤ 22
Reflexions collected	55527	60546	46420	46943
Reflexions unique	6692 [<i>R</i> _{int} = 0.0642]	13810 [<i>R</i> _{int} = 0.0681]	9951 [<i>R</i> _{int} = 0.1509]	10063 [<i>R</i> _{int} = 0.0914]
Refinement method	Full-matrix least-squares on <i>F</i> ²	Full-matrix least-squares on <i>F</i> ²	Full-matrix least-squares on <i>F</i> ²	Full-matrix least-squares on <i>F</i> ²
Data/Restraints/Parameters	6692/0/250	13810/0/473	9951/18/478	10063/90/530
Absorption correction	Empirical	Empirical	Empirical	Empirical
Goodness-of-fit on <i>F</i> ² (<i>GOF</i>)	1.098	0.853	0.910	0.816
Final <i>R</i> indices [<i>I</i> > 2σ(<i>I</i>)]	<i>R</i> _I = 0.0399, <i>wR</i> ₂ = 0.1148	<i>R</i> _I = 0.0689, <i>wR</i> ₂ = 0.1818	<i>R</i> _I = 0.0622, <i>wR</i> ₂ = 0.1536	<i>R</i> _I = 0.0382, <i>wR</i> ₂ = 0.0818
<i>R</i> indices (all data)	<i>R</i> _I = 0.0839, <i>wR</i> ₂ = 0.1417	<i>R</i> _I = 0.1433, <i>wR</i> ₂ = 0.2063	<i>R</i> _I = 0.1446, <i>wR</i> ₂ = 0.2094	<i>R</i> _I = 0.0925, <i>wR</i> ₂ = 0.1253
Largest difference peak and hole (e Å ⁻³)	2.616 and -0.788	2.427 and -1.844	1.704 and -1.296	0.933 and -0.812

Table S1 (continued).

Crystal parameters	5	6	7	8
Empirical formula	C ₅₂ H ₈₄ RuZn ₁₀	C ₄₄ H ₇₂ RuZn ₈	C ₃₆ RhZn ₉	C ₄₄ H ₇₂ GaRhZn ₇
Molecular weight [g/mol]	1463.96	1225.05	1123.60	1231.24
Temperature [K]	113(2)	105(2)	113(2)	113(2)
Wavelength Mo- K_{α} [Å]	0.71073	0.71073	0.71073	0.71073
Crystal size [mm]	0.25 × 0.25 × 0.15	0.2 × 0.2 × 0.1	0.3 × 0.25 × 0.15	0.2 × 0.15 × 0.1
Crystal system	Monoclinic	Monoclinic	Orthorhombic	Monoclinic
space group	P2(1)/n	P2(1)/c	Pmc2(1)	P2(1)/c
a (Å)	17.2946(11)	12.5205(4)	19.5840(7)	16.9525(7)
b (Å)	17.390(2)	23.2905(7)	12.5029(5)	16.5830(4)
c (Å)	18.725(2)	22.5218(8)	17.3314(6)	23.4379(9)
α (°)	90	90	90	90
β (°)	94.516(7)	123.522(2)	90	110.783(4)
γ (°)	90	90	90	90
Cell volume [Å ³]	5614.4(10)	5475.2(3)	4243.7(3)	6160.2(4)
Z	4	4	4	4
Density ρ_{calc} [g cm ⁻³]	1.732	1.486	1.759	1.328
Absorption coefficient μ [mm ⁻¹]	4.481	3.731	5.386	3.387
$F(000)$	2960	2480	2124	2488
θ range for data collection [°]	2.58 - 27.74	2.63 - 27.63	3.12 - 27.65	2.77 - 27.60
Index ranges	-22 ≤ h ≤ 22, -22 ≤ k ≤ 17, -24 ≤ l ≤ 23	-15 ≤ h ≤ 16, -30 ≤ k ≤ 21, -29 ≤ l ≤ 29	-25 ≤ h ≤ 25, -16 ≤ k ≤ 16, -16 ≤ l ≤ 22	-19 ≤ h ≤ 22, -17 ≤ k ≤ 21, -30 ≤ l ≤ 30
Reflexions collected	57352	55612	43385	62241
Reflexions unique	13125 [$R_{\text{int}} = 0.1273$]	12696 [$R_{\text{int}} = 0.1070$]	8613 [$R_{\text{int}} = 0.1004$]	14195 [$R_{\text{int}} = 0.0529$]
Refinement method	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2
Data/Restraints/Parameters	13125/0/285	12696/0/502	8613/13/442	14195/0/478
Absorption correction	Empirical	Empirical	Empirical	Empirical
Goodness-of-fit on F^2 (GOF)	0.832	0.968	0.782	0.882
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0468$, $wR_2 = 0.0839$	$R_1 = 0.0533$, $wR_2 = 0.1456$	$R_1 = 0.0401$, $wR_2 = 0.0625$	$R_1 = 0.0383$, $wR_2 = 0.1055$
R indices (all data)	$R_1 = 0.1580$, $wR_2 = 0.1084$	$R_1 = 0.0951$, $wR_2 = 0.1559$	$R_1 = 0.0902$, $wR_2 = 0.0700$	$R_1 = 0.0720$, $wR_2 = 0.1100$
Largest difference peak and hole (e Å ⁻³)	1.387 and -0.910	3.188 and -1.302	1.774 and -1.287	1.920 and -0.695

Table S1 (continued).

Crystal parameters	9	10	11	12
Empirical formula	C ₄₄ H ₇₂ NiZn ₈	C ₄₄ H ₇₂ PdZn ₈	C ₄₄ H ₆₉ PtZn ₈	C ₅₂ H ₉₀ PtZn ₈
Molecular weight [g/mol]	1182.69	1230.38	1319.07	1449.29
Temperature [K]	113(2)	113(2)	113(2)	113(2)
Wavelength Mo- K_{α} [Å]	0.71073	0.71073	0.71073	0.71073
Crystal size [mm]	0.15 × 0.15 × 0.1	0.25 × 0.15 × 0.1	0.40 × 0.35 × 0.2	0.25 × 0.15 × 0.15
Crystal system	Monoclinic	Monoclinic	Monoclinic	Orthorhombic
space group	C2/c	P2(1)/n	P2(1)/n	Pbcn
a (Å)	29.9453(13)	12.6106(3)	12.5896(12)	17.2256(3)
b (Å)	20.8547(6)	23.3762(9)	23.1888(8)	17.7766(3)
c (Å)	22.1767(7)	18.9914(7)	19.066(4)	18.5676(3)
α (°)	90	90	90	90
β (°)	126.644(3)	93.411(3)	93.262(11)	90
γ (°)	90	90	90	90
Cell volume [Å ³]	11112.2(7)	5588.5(3)	5557.0(12)	5685.63(17)
Z	4	4	4	4
Density ρ_{calc} [g cm ⁻³]	1.414	1.462	1.573	1.693
Absorption coefficient μ [mm ⁻¹]	3.744	3.707	5.919	5.795
$F(000)$	4832	2488	2604	2912
θ range for data collection [°]	2.59 - 27.71	2.57 - 27.56	2.58 - 27.65	2.80 - 27.61
Index ranges	-38 ≤ h ≤ 24, -27 ≤ k ≤ 27, -28 ≤ l ≤ 28	-12 ≤ h ≤ 16, -30 ≤ k ≤ 28, -24 ≤ l ≤ 24	-16 ≤ h ≤ 15, -21 ≤ k ≤ 30, -24 ≤ l ≤ 24	-19 ≤ h ≤ 22, -23 ≤ k ≤ 23, -24 ≤ l ≤ 23
Reflexions collected	40792	56302	56261	55459
Reflexions unique	12862 [$R_{\text{int}} = 0.0669$]	12882 [$R_{\text{int}} = 0.0956$]	12875 [$R_{\text{int}} = 0.0459$]	6591 [$R_{\text{int}} = 0.0383$]
Refinement method	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2
Data/Restraints/Parameters	12862/0/478	12882/0/478	12875/0/483	6591/0/281
Absorption correction	Empirical	Empirical	Empirical	Empirical
Goodness-of-fit on F^2 (GOF)	0.780	0.939	0.994	0.949
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0372$, $wR_2 = 0.0907$	$R_1 = 0.0433$, $wR_2 = 0.1248$	$R_1 = 0.0455$, $wR_2 = 0.1266$	$R_1 = 0.0234$, $wR_2 = 0.0532$
R indices (all data)	$R_1 = 0.0819$, $wR_2 = 0.1001$	$R_1 = 0.0709$, $wR_2 = 0.1410$	$R_1 = 0.0787$, $wR_2 = 0.1384$	$R_1 = 0.0435$, $wR_2 = 0.0570$
Largest difference peak and hole (e Å ⁻³)	0.944 and -0.523	1.294 and -1.590	2.146 and -2.873	0.973 and -0.717

In the refinement of structures **3**, **6**, **8**, **9**, **10**, **11** Platon Squeeze was applied to remove unrefineable disordered solvent molecules.

Table S2. Calculated metal-metal bond length of the model compounds **1H**, **4H**, **5H**, **8H** - **11H** where the organic moieties are replaced by hydrogen atoms.

compound			M-Zn	M-Ga	Zn-Zn	Ga-Ga	Ga-Zn
Mo(ZnH) ₁₂	1H	RI-BP86/def2-TZVPP (<i>I_h</i>)	2.684	-	2.822	-	-
Mo(GaH) ₄ (ZnH) ₄	3H	RI-BP86/def2-TZVPP (<i>D₂</i>)	2.586	2.417	3.143	3.279	2.928-3.116
Mo(GaH) ₂ (ZnH) ₈	4H	RI-BP86/def2-TZVPP (<i>C_s</i>)	2.627-2.651	2.434; 2.440	2.763-3.124	-	2.824-3.000
Ru(ZnH) ₁₀	5H	RI-BP86/def2-TZVPP (<i>D_{4d}</i>)	2.541; 2.584	-	2.740; 2.798	-	-
Rh(ZnH) ₉	7H	RI-BP86/def2-TZVPP (<i>D_{3h}</i>)	2.497; 2.516	-	2.853; 3.044	-	-
Rh(GaH)(ZnH) ₇	8H	RI-BP86/def2-TZVPP (<i>C_s</i>)	2.481-2.486	2.308	2.835-3.142	-	2.989; 3.244
Ni(ZnH) ₈	9H	RI-BP86/def2-TZVPP (<i>D_{4d}</i>)	2.374	-	2.821; 2.993	-	-
Pd(ZnH) ₈	10H	RI-BP86/def2-TZVPP (<i>D_{4d}</i>)	2.482	-	2.936; 3.149	-	-
Pt(ZnH) ₈	11H	RI-BP86/def2-TZVPP (<i>D_{4d}</i>)	2.496	-	2.951; 3.172	-	-

Table S3: Results of the EDA analyses of Ru(ZnH)₁₀ (**5H**) at BP86/TZ2P+ with different electronic states of the fragments. Energies in kcal/mol.

	Ru ²⁺ (s ¹ d ⁵) + (ZnH) ₁₀ ²⁻ (septet)	Ru(s ⁰ d ⁸) ^[a] + (ZnH) ₁₀ (triplett)	Ru(s ⁰ d ⁸) ^[b] + (ZnH) ₁₀ (singlet)
ΔE_{int}	-956.2	-342.3	-392.6
ΔE_{Pauli}	448.1	511.3	477.4
$\Delta E_{\text{elstat}}^{[c]}$	-518.0 (36.9%)	-542.0 (63.5%)	-531.1 (61.1%)
$\Delta E_{\text{Orb}}^{[c]}$	-886.3 (63.1%)	-311.5 (36.5%)	-338.9 (39.0%)
$\Delta E_{\text{Orb}} (a_1)(s, d_{z^2})^{[d]}$	-212.2 (24.0%)	-78.6 (25.2%)	-94.1 (27.8%)
$\Delta E_{\text{Orb}} (a_2)^{[d]}$	-0.3 (<0.1%)	-0.0 (0.0%)	-0.0 (0.0%)
$\Delta E_{\text{Orb}} (b_1)^{[d]}$	-0.5 (0.1%)	-0.0 (0.0%)	-0.0 (0.0%)
$\Delta E_{\text{Orb}} (b_2)(p_z)^{[d]}$	-35.2 (4.0%)	-4.2 (1.4%)	-5.2 (1.5%)
$\Delta E_{\text{Orb}} (e_1)(p_x, p_y)^{[d]}$	-65.4 (7.4%)	-7.0 (2.2%)	-7.7 (2.3%)
$\Delta E_{\text{Orb}} (e_2)(d_{x^2-y^2}, d_{xy})^{[d]}$	-291.7 (32.9%)	-126.0 (40.5%)	-115.4 (34.1%)
$\Delta E_{\text{Orb}} (e_3)(d_{xz}, d_{yz})^{[d]}$	-280.9 (31.7%)	-95.6 (30.7%)	-116.5 (34.4%)

^[a] In Ru(s⁰ d⁸) (triplet) the valence d_{xz} and d_{yz} (e₃) AOs are singly occupied.

^[b] In Ru(s⁰ d⁸) (singlet) the valence d_{z²} AO is vacant.

^[c] The percentage values in parentheses give the contribution to the total attractive interactions $\Delta E_{\text{elstat}} + \Delta E_{\text{orb}}$.

^[d] The percentage values in parentheses give the contribution to the total orbital interactions ΔE_{orb} .

Table S4. EDA results at BP86/TZ2P for $[M(\text{ZnH})_8]$ ($M = \text{Ni, Pd, Pt}$) with the fragments M ($s^0 d^{10}$) and $(\text{ZnH})_{10}$ in the singlet state. Energy values in kcal/mol.

	Ni	Pd	Pt
ΔE_{int}	-246.3	-201.2	-279.0
ΔE_{Pauli}	208.1	402.4	486.0
$\Delta E_{\text{elstat}}^{[a]}$	-310.8 (68.4%)	-467.7 (77.5%)	-583.4 (76.3%)
$\Delta E_{\text{orb}}^{[a]}$	-143.6 (31.6%)	-135.8 (22.5%)	-181.6 (23.7%)
$\Delta E(a_1) (s, d_{z^2})^{[b]}$	-23.4 (16.3%)	-21.7 (16.0%)	-43.8 (24.1%)
$\Delta E(a_2)^{[b]}$	0.0 (0.0%)	0.0 (0.0%)	0.0 (0.0%)
$\Delta E(b_1)^{[b]}$	0.0 (0.0%)	0.0 (0.0%)	0.0 (0.0%)
$\Delta E(b_2) (p_z)^{[b]}$	-5.9 (4.1%)	-5.4 (4.0%)	-8.2 (4.5%)
$\Delta E(e_1) (p_x, p_y)^{[b]}$	-12.0 (8.3%)	-11.0 (8.1%)	-16.7 (9.2%)
$\Delta E(e_2) (d_{xy}, d_{x^2-y^2})^{[b]}$	-46.5 (32.4%)	-43.6 (32.1%)	-51.2 (28.2%)
$\Delta E(e_3) (d_{xz}, d_{yz})^{[b]}$	-55.8 (38.9%)	-54.1 (39.8%)	-61.7 (34.0%)

[a] Percentage values in parentheses give the contributions to the total attractive energy $\Delta E_{\text{elstat}} + \Delta E_{\text{orb}}$.

[b] Percentage values in parentheses give the contributions to the total orbital energy ΔE_{orb} .

Table S5: Results of the EDA analyses of Rh(ZnH)₉ in D_{3h} at BP86/TZ2P+ with different electronic states of the fragments. Energies in kcal/mol.

	Rh ⁻ (s ⁰ d ¹⁰) + (ZnH) ₉ ⁺ (singlett)	Rh (s ⁰ d ⁹) + (ZnH) ₉ (doublett)
ΔE_{int}	-457.4	-294.5
ΔE_{Pauli}	598.4	465.1
ΔE_{elstat} ^[a]	-883.0 (83.6%)	-518.5 (68.3%)
ΔE_{orb} ^[a]	-172.8 (16.4%)	-241.0 (31.7%)
$\Delta E(a_1')$ (s, d _{z2}) ^[b]	-28.8 (16.7%)	-62.0 (25.7%)
$\Delta E(a_2')$ ^[b]	-0.2 (0.1%)	-0.3 (0.1%)
$\Delta E(e_1')$ (p _x , p _y , d _{xy} , d _{x2-y2}) ^[b]	-65.9 (38.2%)	-85.5 (35.5%)
$\Delta E(a_1'')$ ^[b]	-0.1 (0.0%)	-0.0 (0.0%)
$\Delta E(a_2'')$ (p _z) ^[b]	1.0 (-0.6%)	-5.2 (2.2%)
$\Delta E(e_1'')$ (d _{xz} , d _{yz}) ^[b]	-78.8 (45.6%)	-88.1 (36.6%)

[a] Percentage values in parentheses give the contributions to the total attractive energie $\Delta E_{\text{elstat}} + \Delta E_{\text{orb}}$.

[b] Percentage values in parentheses give the contributions to the total orbital energy ΔE_{or}

Table S6: Calculated (BP86/TZ2P+) Hirshfeld atomic partial charges of the model compounds M(ZnH)_n.

compound	M	Zn	H
Mo(ZnH) ₁₂	-0.228	0.159	-0.140
Ru(ZnH) ₁₀	-0.146	0.176	-0.157;-0.163
Rh(ZnH) ₉	-0.172	0.180;0.184	-0.163;-0.164
Ni(ZnH) ₈	-0.121	0.178	-0.163
Pd(ZnH) ₈	0.096	0.154	-0.166
Pt(ZnH) ₈	-0.117	0.179	-0.165

General comments on the terms used to describe different types of covalent bonding.

In view of the non-trivial bonding situation of the transition metal compounds which are reported here we want to explain the terms which we are using for the different types of bonding. We distinguish between an electron sharing bond A-B where bond formation formally takes place between two open-shell fragments A and B which contribute with one electron each to the covalent bond and a donor-acceptor bond A-B where closed-shell fragments bind in a way which is best described with the familiar Dewar-Chatt-Duncanson (DCD) model of donation and backdonation.⁵⁹ Furthermore, it is obvious that redox processes are involved in the reactions. For example, ZnMe_2 is clearly a Zn(II) compound but the bonding situation in $[\text{Mo}(\text{ZnCp}^*)_3(\text{ZnMe})_9]$ (**1**) shows that six of the twelve valence electrons of the ZnR ligands are used for electron-sharing Mo-Zn bonds while six are used for delocalized Zn-Zn bonds. Taking into account the rather different electronegativities (Pauling scale) of Mo (2.2) and Zn (1.7) formal oxidation states of -6 for Mo and +1.5 for Zn are derived. However, using Allred Rochow electronegativities with Mo (1.3) and Zn (1.7) a different result of +6 for Mo and +0.5 for Zn may be obtained. Obviously, the situation is ambiguous and becomes especially complicated when the metal-ligand bonding comes from a mixture of donor-acceptor and electron sharing interactions and when GaR and ZnR ligands are involved.⁶⁰ We agree with one referee who argued that the oxidation state concept breaks down for our molecules. Thus, we will refrain from the assignment of formal oxidation states for the metal atoms in the compounds **1-12**. Rather, the bonding situation can be well described when the molecular orbitals are analyzed with charge and energy decomposition methods which shall be discussed in the following.

Additional References

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